

SAQI / AgriBot

Hardware Datasheet - GPIO Pin Map, Wiring & Code

Raspberry Pi agricultural robot: differential drive (2x BTS7960), relay-driven water pump, WS2812 8-LED status panel, HC-SR04 ultrasonic sensor, and an AXIS PTZ network camera.

Single source of truth: every GPIO pin and wiring parameter in this document is defined in `Backend/components/config.py`. Edit pins there only - the controller modules import from it.

All pin numbers use **BCM GPIO** numbering (not physical header position).

Document generated from `config.py` (authoritative).

1. System Overview

Subsystem	Device	Interface
Drive	2x BTS7960 H-bridge + DC gear motors	GPIO PWM (gpiozero Robot)
Water pump	Relay module + pump	GPIO (active-LOW)
Status LEDs	WS2812 / NeoPixel x8	GPIO18 data (PWM)
Ultrasonic	HC-SR04	GPIO trigger + echo (gpiozero)
Camera	AXIS PTZ (network)	HTTP / MJPEG - NO GPIO

The camera is a network device (no GPIO wiring). Everything else is wired to the Pi 40-pin header as detailed below.

2. Master GPIO Pin Map

BCM	Subsystem	Function	Direction	config.py
GPIO12	Left motor	RPWM (forward PWM)	OUT	MOTOR_LEFT_FWD_PIN
GPIO13	Left motor	LPWM (backward PWM)	OUT	MOTOR_LEFT_BWD_PIN
GPIO5	Left motor	Enable A (R_EN)	OUT	MOTOR_LEFT_EN_PIN
GPIO17	Left motor	Enable B (L_EN)	OUT	MOTOR_LEFT_EN_PIN1
GPIO22	Right motor	RPWM (forward PWM)	OUT	MOTOR_RIGHT_FWD_PIN
GPIO23	Right motor	LPWM (backward PWM)	OUT	MOTOR_RIGHT_BWD_PIN
GPIO6	Right motor	Enable A (R_EN)	OUT	MOTOR_RIGHT_EN_PIN
GPIO27	Right motor	Enable B (L_EN)	OUT	MOTOR_RIGHT_EN_PIN1
GPIO25	Water pump	Relay IN (active-LOW)	OUT	PUMP_PIN
GPIO18	Status LEDs	WS2812 data in	OUT	LED_PIN
GPIO20	Ultrasonic	HC-SR04 TRIG	OUT	ULTRASONIC_TRIG_PIN
GPIO21	Ultrasonic	HC-SR04 ECHO (divider!)	IN	ULTRASONIC_ECHO_PIN

Parameters: motor PWM = 1000 Hz (MOTOR_PWM_FREQUENCY, anti-whine) - pump active_high = False (PUMP_ACTIVE_HIGH) - LEDs: 8 count / brightness 0.20 (LED_COUNT, LED_BRIGHTNESS) - ultrasonic max 200 cm, obstacle threshold 30 cm (ULTRASONIC_MAX_CM, ULTRASONIC_OBSTACLE_CM).

CAUTION: Pick free GPIOs only. The defaults above do not conflict. GPIO18 is also the hardware-PWM line used by WS2812 - do not reuse it.

3. Motor Drivers - 2x BTS7960

Differential drive: one BTS7960 H-bridge per side. gpiozero drives each side as a (forward, backward) PWM pair; both BTS7960 enable pins per side are held HIGH at startup so both half-bridges are active.

3.1 Per-driver wiring

BTS7960 pin	LEFT driver	RIGHT driver	Notes
RPWM	Pi GPIO12	Pi GPIO22	Forward PWM input
LPWM	Pi GPIO13	Pi GPIO23	Backward PWM input
R_EN	Pi GPIO5	Pi GPIO6	Enable - driven HIGH on startup
L_EN	Pi GPIO17	Pi GPIO27	Enable - driven HIGH on startup
VCC	Pi 3V3	Pi 3V3	Logic supply for inputs (per board)
GND	Pi GND	Pi GND	MUST be common with Pi ground
B+ / B-	Motor battery + / -	Motor battery + / -	Motor power (e.g. 12 V pack)
M+ / M-	Left motor	Right motor	H-bridge output to motor

CAUTION: Never power the motors from the Pi 5V rail. Use a separate battery for B+/B- and share ONLY ground with the Pi.

3.2 Behavior

- forward()/backward() set both sides equal; left()/right() spin in place (internally speed x2, capped at 1.0).
- PWM frequency forced to 1000 Hz to stop audible whine.
- On close(): motors stop, then all four enable pins are driven LOW (drivers asleep).

4. Water Pump - Relay

Relay pin	Connect to	Notes
IN / SIG	Pi GPIO25	Active-LOW: LOW = pump ON
VCC	Pi 5V (or ext)	Relay coil supply
GND	Pi GND	Common ground
COM / NO	Pump power loop	Switch the pump's own supply, not the Pi

Configured active-LOW (PUMP_ACTIVE_HIGH = False) and initialised OFF, so the pump stays off at boot regardless of GPIO float.

CAUTION: Drive the pump from its own supply through the relay contacts. Add a flyback diode across an inductive pump if the relay board lacks one.

5. Status LEDs - WS2812 / NeoPixel (x8)

Strip pin	Connect to	Notes
DIN	Pi GPIO18	Serial data (PWM). First pixel's data input
5V / VCC	5V supply	WS2812 runs at 5V; size supply for 8 px
GND	Pi GND + supply GND	Common ground REQUIRED

8-LED status panel (index -> meaning), auto-driven by system events:

#	LED	Meaning / colors
1	Manual/Auto	Manual = red, Auto = green, init = orange blink
2	Wifi	Ready = green, error = red, init = orange blink
3	Camera	Ready = green, error = red, init = orange blink
4	Ultrasonic	Blue ON when obstacle < 30 cm
5	Moving	Blue ON while motors run
6	Detect	Blue ON when YOLO sees a plant
7	Watering	Green ON while pump runs
8	Error	Red blink on failure

CAUTION: WS2812 on GPIO18 uses `rpi_ws281x` which typically needs ROOT. Run the server with `sudo`, or the LED panel self-disables (the robot still runs).

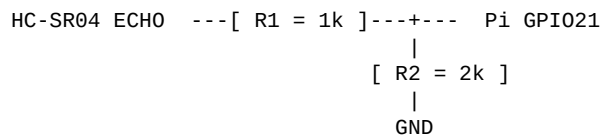
Data-level note: WS2812 expects a ~5V data signal; the Pi GPIO is 3.3V. A short strip often works directly, but a 3.3V->5V level shifter on DIN is recommended for reliability.

6. Ultrasonic - HC-SR04

HC-SR04 pin	Connect to	Notes
VCC	Pi 5V	HC-SR04 needs 5V to work reliably
TRIG	Pi GPIO20	3.3V trigger from Pi is accepted
ECHO	Pi GPIO21 via divider	ECHO is 5V - MUST drop to 3.3V
GND	Pi GND	Common ground

CAUTION: HC-SR04 ECHO outputs 5V. Connecting it directly to a Pi GPIO can damage the Pi. Use a resistor divider to bring it to ~3.3V.

6.1 ECHO voltage divider



$$V_{out} = 5V * R2 / (R1 + R2) = 5 * 2k / (1k + 2k) = 3.33V \quad (\text{safe})$$

Implemented with `gpiozero.DistanceSensor(echo=GPIO21, trigger=GPIO20, max_distance=2.0 m)`. A background poll (~5 Hz) updates the reading and drives LED #4. Reading + LED only - it does not change autonomous navigation.

7. Power & Grounding

- **Common ground:** Pi GND, motor battery GND, LED supply GND and the relay/sensor GND must all be tied together.
- **Separate rails:** motors from their own battery (BTS7960 B+/B-); do not back-feed the Pi.
- **Pi supply:** use an adequate 5V supply; brown-outs during motor surge cause resets - power the Pi independently of the motor pack.

8. Code Examples

8.1 Single source of truth - components/config.py

```
MOTOR_LEFT_FWD_PIN = 12 # left RPWM
MOTOR_LEFT_BWD_PIN = 13 # left LPWM
MOTOR_LEFT_EN_PIN  = 5  # left enable A
MOTOR_LEFT_EN_PIN1 = 17 # left enable B
MOTOR_RIGHT_FWD_PIN = 22 # right RPWM
MOTOR_RIGHT_BWD_PIN = 23 # right LPWM
MOTOR_RIGHT_EN_PIN  = 6  # right enable A
MOTOR_RIGHT_EN_PIN1 = 27 # right enable B
MOTOR_PWM_FREQUENCY = 1000 # Hz

PUMP_PIN          = 25
PUMP_ACTIVE_HIGH  = False

LED_PIN           = 18
LED_COUNT         = 8
LED_BRIGHTNESS    = 0.20

ULTRASONIC_TRIG_PIN = 20
ULTRASONIC_ECHO_PIN = 21
ULTRASONIC_MAX_CM   = 200
ULTRASONIC_OBSTACLE_CM = 30
```

8.2 Motor

```

from components.motor import MotorController
m = MotorController()      # pins come from config.py
m.forward(0.8)             # both sides forward at 80%
m.left(0.6)               # spin left in place
m.stop()
m.close()                 # stop + disable drivers

```

8.3 Water pump

```

from components.waterpump import WaterPumpController
pump = WaterPumpController() # GPIO25, active-LOW
pump.on(); pump.off()
print(pump.is_on)
pump.close()

```

8.4 Ultrasonic

```

from components.ultrasonic import UltrasonicSensor
u = UltrasonicSensor()      # gpiozero.DistanceSensor (HC-SR04)
u.start()                  # background poll -> drives LED #4
print(u.distance_cm(), u.is_obstacle())
u.close()

```

8.5 Status LEDs

```

from components.leds import SAQI_LEDS, LEDState
SAQI_LEDS.initialize()     # safe no-op if hardware/libs absent
SAQI_LEDS.mode(LEDState.AUTO)
SAQI_LEDS.wifi(LEDState.READY)
SAQI_LEDS.camera(LEDState.READY)
SAQI_LEDS.moving(True)
SAQI_LEDS.watering(True)
SAQI_LEDS.error(True)
SAQI_LEDS.all_off()

```

8.6 Run

```

# Backend only
cd Backend && ./start.sh

# Whole system (Backend + Frontend)
cd .. && ./start.sh

# WS2812 LEDs need root:
sudo Backend/venv/bin/python -m uvicorn main:app --host 0.0.0.0 --port 8000

```